Demo: Rubens' Tube

See, measure, and verify a standing wave safely

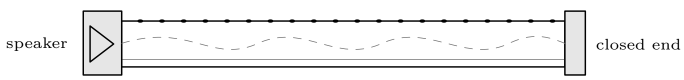
An observation notebook for a professionally assembled flame demonstration

Lesson Demo · one supervised session · **adult demonstrator owns every flame and gas control**

DEMONSTRATION APPARATUS — NOT A STUDENT GAS PROJECT

Use only outdoors in a cleared demonstration area, with a trained adult operating a sound, commercially made or professionally assembled metal apparatus. Follow its written instructions and fuel supplier's rules. Students do not build, connect, fill, light, adjust, touch, or approach it. No improvised pressure parts, no indoor or garage operation, and no attempt to make flames larger. Extinguisher and fuel shutoff stay with the operator. Any leak, damage, unstable flame, unexpected sound, or loss of supervision means: shut off remotely if it is safe to do so, withdraw, and do not relight.

First make a prediction



the drawing is a model of pressure inside — not a build plan

Predict first

When the operator plays one steady tone, will the row stay level, flicker together, travel from end to end, or settle into fixed tall and short regions? Sketch your prediction before the apparatus is lit.

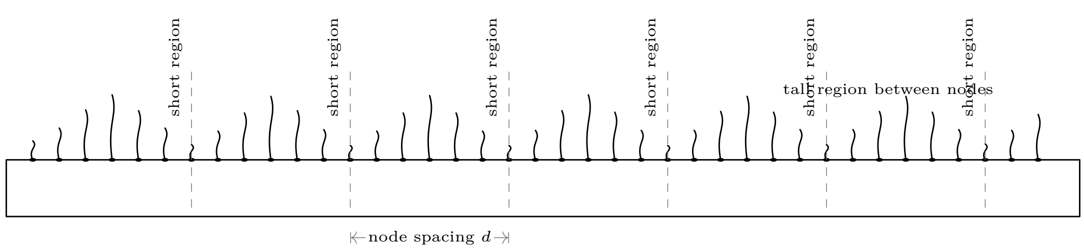
Question 1 What observation would distinguish a travelling pattern from a standing one?

Question 2 If two neighbouring short regions are nodes, where should the tallest region lie?

Question 3 Which is evidence: one striking photograph, or the same fixed pattern seen repeatedly at the same tone? Why?

Control Record one steady-tone observation and one "speaker silent" observation. The difference is more informative than spectacle alone.

The wave the flames reveal



Read the pattern cautiously. Reflection and interference can make fixed pressure regions, but drafts, warming, geometry, and hole differences also affect height. Use repeated short-region centres as qualitative markers. The safe aim is a modest, stable pattern; flame size is never a score.

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Observe in order — do not change two things at once

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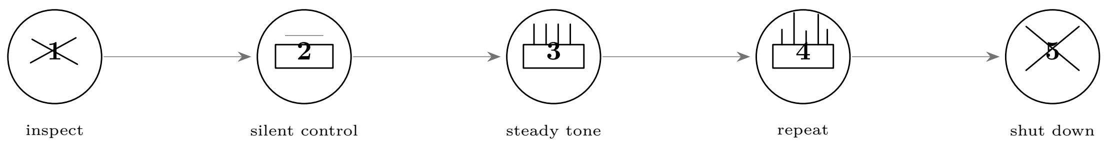
Lesson Sequence · predict, control, observe, repeat · **operator calls every hold point**

A controlled observation sequence



1. **Cold hold point.** Operator verifies the written apparatus checklist, clear outdoor area, stable supports, intact barrier, weather, extinguisher, and shutoff access. Students remain behind the marked line.
2. **Silent control.** With no speaker drive, observe only from the assigned position. Record whether the row is acceptably stable; do not diagnose or correct apparatus faults while it is operating.
3. **One steady tone.** The operator selects a documented, permitted setting. Wait for a stable pattern; observers mark short-region centres without approaching the tube.
4. **Repeat.** Shut down as the apparatus instructions require. After the operator authorizes another observation, repeat the same setting. Agreement is the test; do not chase a prettier pattern.
5. **Final shutdown.** Operator closes the fuel supply, follows the manufacturer's shutdown procedure, maintains the exclusion zone through cool-down, and records completion.

Fault gate — stop means stop

If the pre-use checklist fails, the row is irregular before sound is applied, the pattern changes without a commanded change, flame appears anywhere except the intended row, wind disturbs the demonstration, a support moves, the speaker seal changes, or any person crosses the boundary: the run fails. The operator shuts down if safe, isolates the apparatus, and has the defective part remade or professionally repaired. Never patch, tighten, drill, or adjust a fueled or hot apparatus. A failed run produces no measurement.

Observation record

Run	Tone / ID	Short-region positions from fixed scale	Stable?	Keep / reject
control	silent			
1A				
1B repeat				
2A				
2B repeat				

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Measure the record, not the flame

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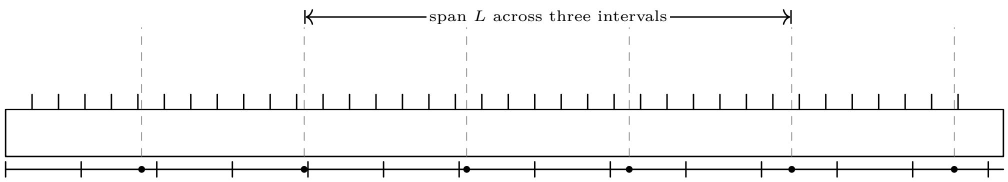
Lesson Measurement · fixed scale and repeated marks · **rejected runs stay out of the arithmetic**

Measure node spacing from a safe distance



fixed scale read by camera or distant observer — nobody measures over the apparatus

Calculation

Choose several consecutive, clearly repeated short regions. Measure the span L from the first centre to the last and count the number N of *intervals*, not marks:

$$d = \frac{L}{N}, \qquad \lambda \approx 2d, \qquad v = f\lambda.$$

The last relation is a model estimate. Label it with the gas, apparatus, tone, date, and uncertainty; do not report it as a universal constant.

Measurement questions

1. Why is a span across several intervals better than one gap?
2. Did repeats place each short region at the same scale mark?
3. Which centre was hardest to locate? How much could it move?
4. Does the silent control contain an equally regular spacing?
5. Would your conclusion survive the largest reasonable marking error?

Measurement record

Run	f / source	first–last	L	intervals N	$d = L/N$	uncertainty / note
repeat						

Accept The cold inspection passed; control was recorded; the pattern remained stable; repeated marks agree within stated reading uncertainty; and every number is traceable to a retained run.

Reject Any safety fault, missing control, moving pattern, ambiguous centres, undocumented setting, or disagreement larger than the stated uncertainty. Cross out the arithmetic but keep the row as evidence of why it failed.

Remake decision A damaged, leaking, distorted, or unstable component is not "good enough for one more run." Tag the apparatus out. A qualified adult decides whether the maker replaces the component or the entire assembly.

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Close the evidence chain

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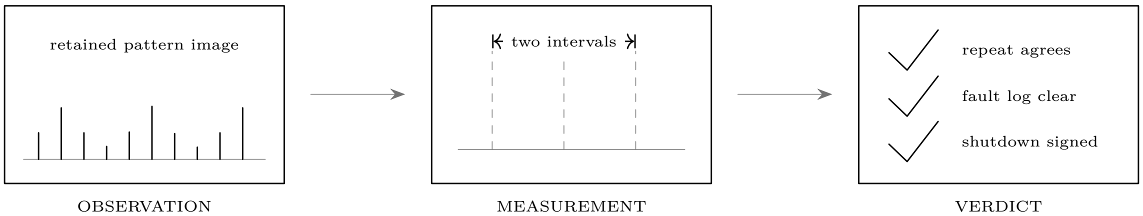
Lesson Final proof · inspection, repeatability, shutdown · **a photograph alone is not proof**

Final verification plate



Inspection and disposition record

Hold point	Evidence / record ID	Operator initials
Cold apparatus checklist passed		
Area, barrier, weather, extinguisher, shutoff passed		
Silent control retained		
Steady-tone run and same-setting repeat retained		
Node centres and uncertainty recorded		
Faults resolved by stop / repair / remake before any later run		
Fuel isolated; apparatus cooled; area released		

Final claim

Complete only after shutdown

At the documented tone, the retained observations ☐ do ☐ do not show fixed short regions that recur at the same positions. The silent control ☐ does ☐ does not show the same regular pattern. Repeated spacing estimates agree within ±________. Therefore the evidence ☐ supports ☐ does not yet support the standing-wave model for this session.

Observer: ____________________ Date: _____________ Adult operator: ____________________

> **Proof:** passed inspection, explicit control, two agreeing observations, traceable measurements, a clear fault record, and confirmed shutdown. **Not proof:** a dramatic flame, one image, selected "best" gaps, or a run after a stop condition. A fixed, repeatable pattern tied to a steady tone supports stationary pressure regions; the record makes that conclusion defensible.